AI EXPLAINED SIMPLY

The AI Chip Map: Design, Make, Machine

Why almost every top AI chip depends on one island, and the three places that make it possible.

The three lines

1. **Design** - Santa Clara, California (USA). Nvidia designs its AI chips here. It has no factory of its own for them.
2. **Manufacturing** - Hsinchu and Tainan (Taiwan). TSMC builds the chips. Santa Clara to Hsinchu is about 10,479 km along the shortest path on the globe.
3. **The machine** - Veldhoven (Netherlands). ASML is the only company that builds the extreme ultraviolet (EUV) lithography machines used for the most advanced chips. Veldhoven to Hsinchu is about 9,475 km.

Key numbers (with sources)

- TSMC held about 70% of global foundry revenue in 2025 (about $122.5 billion in sales), up from 64.4% in 2024 - TrendForce, via Tom's Hardware.
- 7 nm and smaller chips made up 74% of TSMC wafer revenue in Q4 2025 - same source.
- A different firm, Counterpoint, uses a broader "Foundry 2.0" definition and puts TSMC near 38%. The two figures measure different things.

The twist: Arizona

- In October 2025, Nvidia unveiled the first Blackwell wafer made in the USA at TSMC's Phoenix, Arizona fab (PC Gamer).
- It is one wafer on a 5 nm-class process, and it still needs packaging. It is a first step, not a full supply chain.

Glossary

- **Foundry**: a factory that makes chips designed by other companies.
- **Wafer**: the thin silicon disc on which hundreds of chips are printed.
- **EUV lithography**: printing circuits with extreme ultraviolet light, needed for the smallest chip features.

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